



Analog IC Design

CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 – Lab03

Cascode amplifier

Cascode Amplifier

Part 1: Sizing Chart

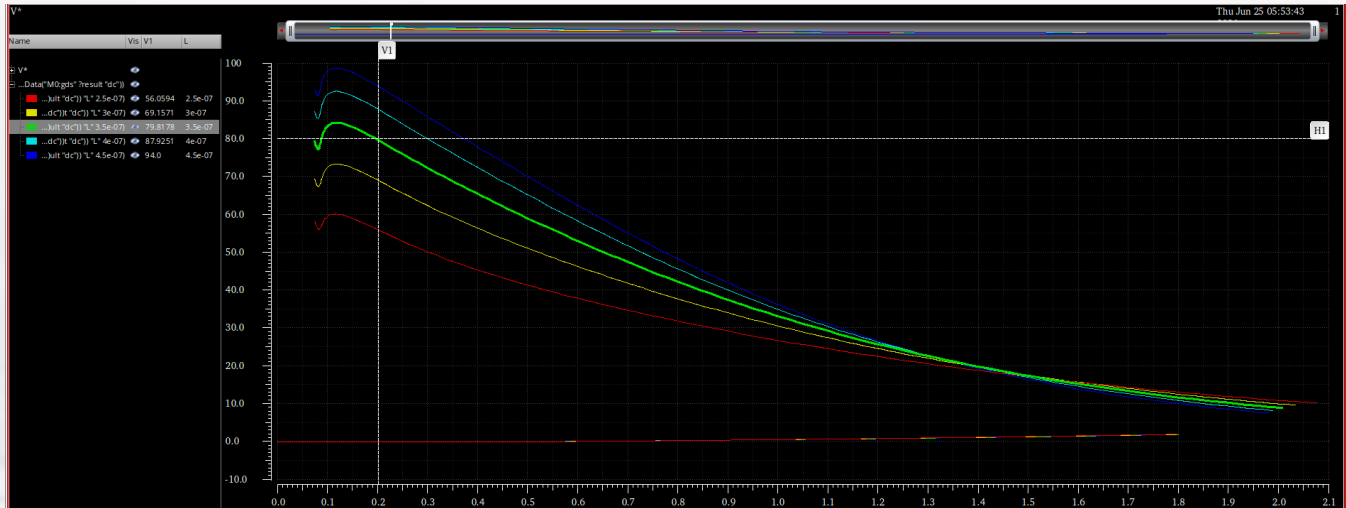
The technology used: 180nm CMOS, So the supply V_{DD} will be 1.8v.

The intrinsic gain $|A_v| \approx g_m r_o = 80$. The bias current $I_D = 50\mu A$.

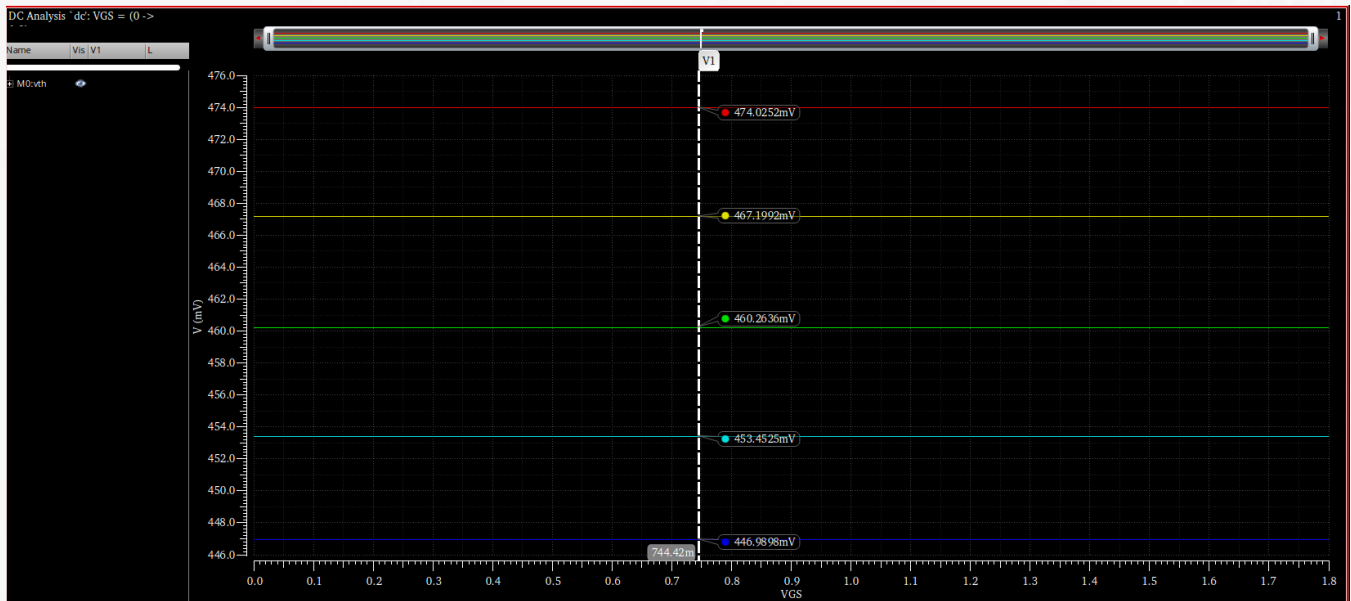
$$V^* = \frac{2I_D}{g_m} = 200mv \text{ so, we expect to get } \rightarrow g_m = \frac{2I_D}{V^*} = \frac{2 \times 50\mu A}{200mv} = 0.5m\Omega^{-1}.$$

$$V_{DS} = \frac{V_{DD}}{2} = \frac{1.8v}{2} = 0.9v.$$

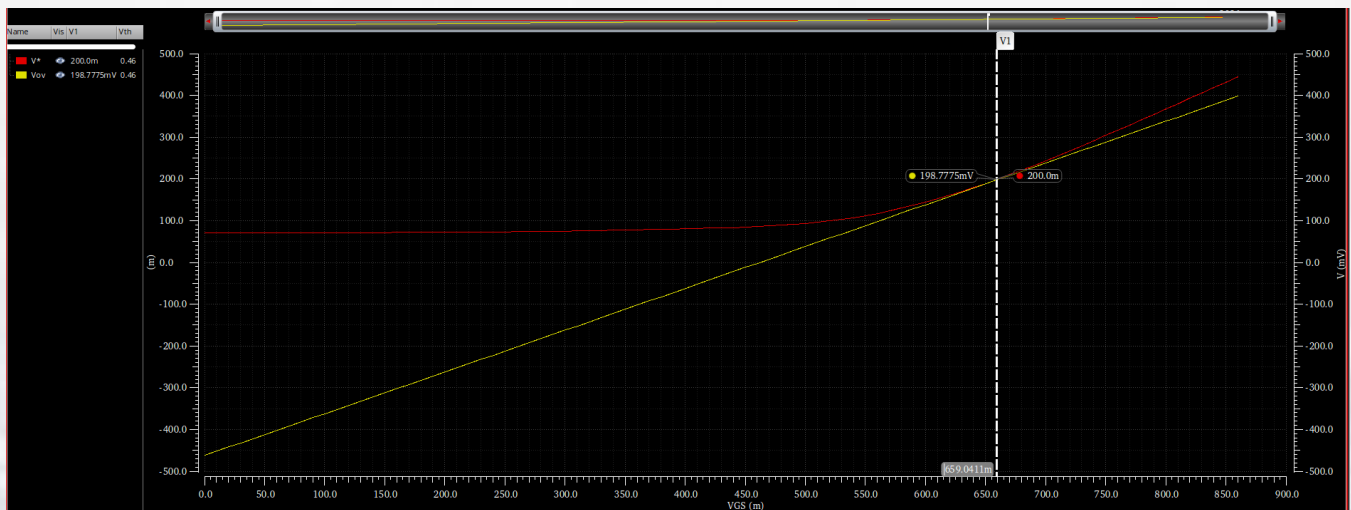
Let the width $W = 10\mu m$. To get the length L that will provide $V^* = 200mv$ and $g_m r_o = 80$, I will make DC analysis and sweep L in a certain range (from 250nm to 450nm with step 50nm), then plotting $g_m r_o$ vs V^* and choose the length of the curve that intersects (or very near) with the point ($g_m r_o = 80$, $V^* = 200mv$) to be the length used in our circuit design.



From the graph above we can see that the length L that can provide our spec is $L \approx 350nm$.

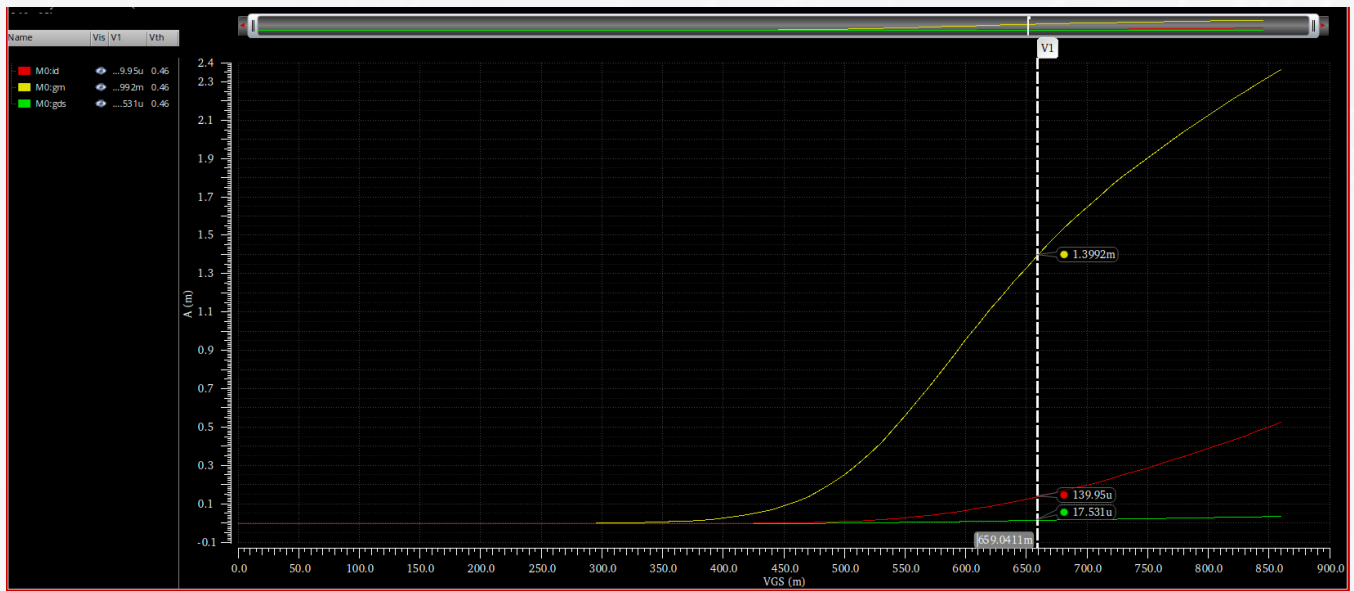


Also, we can get V_{TH} from result browser, $V_{TH} \approx 460.26\text{mV}$ at $L = 350\text{nm}$, So, we will sweep V_{GS} to $V_{TH} + 0.4\text{V} \approx 0.9\text{V}$.



$$\text{Red } V^* = \frac{2I_D}{g_m} \quad \text{Yellow } V_{ov} = V_{GS} - V_{TH}, \text{ applied } w = 10\mu\text{m and } L = 350\text{nm}$$

The corresponding V_{ovQ} and V_{GSQ} to $V^* = 200\text{mV}$ are 198.78mV and 659.041mV respectively.



■ g_m ■ I_D ■ g_{ds}

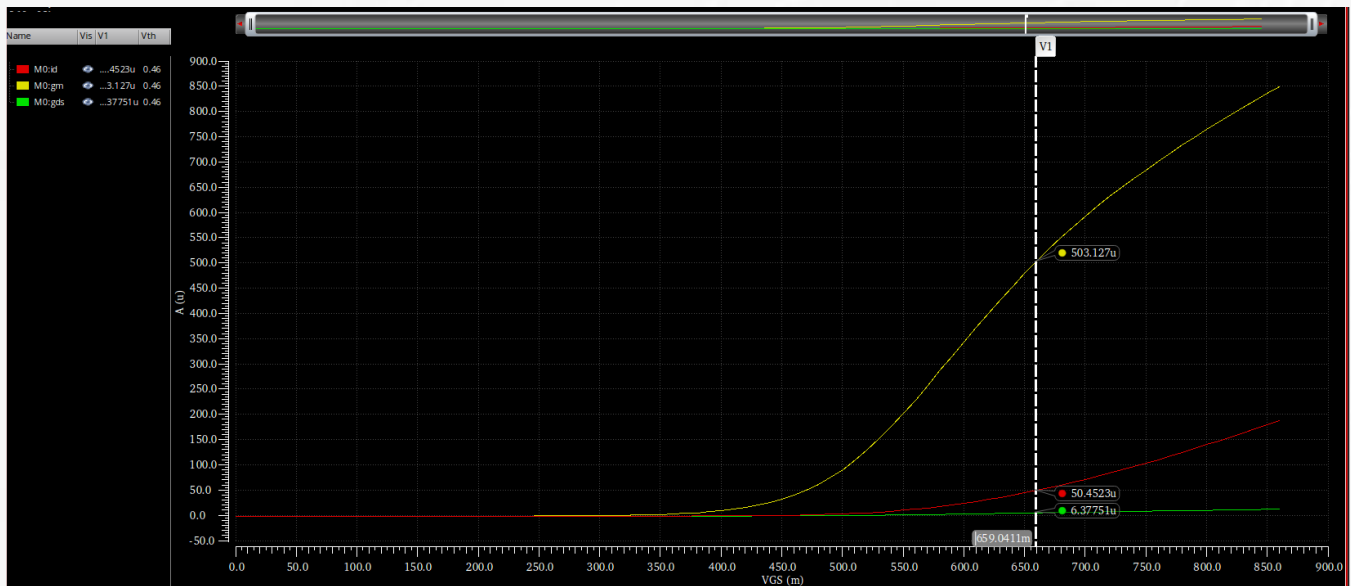
The values of I_D , g_m and g_{ds} at $V_{GSQ} = 659.041 \text{ mV}$ are I_{DX} , g_{mX} and g_{dsX} respectively.

$I_{DX} \approx 140 \mu\text{A}$, $g_{mX} \approx 1.4 \text{ m}\Omega^{-1}$ and $g_{dsX} \approx 17.53 \mu\Omega^{-1}$

Since $I_D \propto w \rightarrow \frac{I_{D1}}{I_{D2}} = \frac{w_1}{w_2} \rightarrow w_2 = w_1 \times \frac{I_{D2}}{I_{D1}}$

Let $w_1 = 10 \mu\text{m} \rightarrow I_{D1} = 140 \mu\text{A}$

the width w_2 that can provide $I_{D2} = 50 \mu\text{A}$ given by $w_2 = w_1 \times \frac{I_{D2}}{I_{D1}} = 10 \mu\text{m} \times \frac{50 \mu\text{A}}{140 \mu\text{A}} \approx 3.57 \mu\text{m}$



The value of I_D after setting the correct width: $50.45\mu\text{m}$

since $g_m \propto w$ and $g_{ds} \propto w \rightarrow \frac{g_{m1}}{g_{m2}} = \frac{W1}{W2}$ and $\frac{g_{ds1}}{g_{ds2}} = \frac{W1}{W2}$

If $w1 = 10\mu\text{m} \rightarrow g_{m1} \approx 1.4\text{m}\Omega^{-1}$ and $g_{ds1} \approx 17.53\mu\Omega^{-1}$

At $w2 = 3.57\mu\text{m}$ we can get g_{m2} and g_{ds2} by:

$$g_{m2} = g_{m1} \times \frac{w2}{w1} = 1.4\text{m}\Omega^{-1} \times \frac{3.57\mu\text{m}}{10\mu\text{m}} \approx 500\mu\Omega^{-1} \quad (\approx \text{as shown in the graph above})$$

$$g_{ds2} = g_{ds1} \times \frac{w2}{w1} = 17.53\mu\Omega^{-1} \times \frac{3.57\mu\text{m}}{10\mu\text{m}} \approx 6.3\mu\Omega^{-1} \quad (\approx \text{as shown in the graph above})$$

double check: $r_o = \frac{1}{g_{ds}} = \frac{1}{6.3\mu\Omega^{-1}} \approx 0.16\text{M}\Omega$

$$|A_v| = g_m r_o = 500\mu\Omega^{-1} \times 0.16\text{M}\Omega = 80$$

$\rightarrow w = 3.57\mu\text{m}$

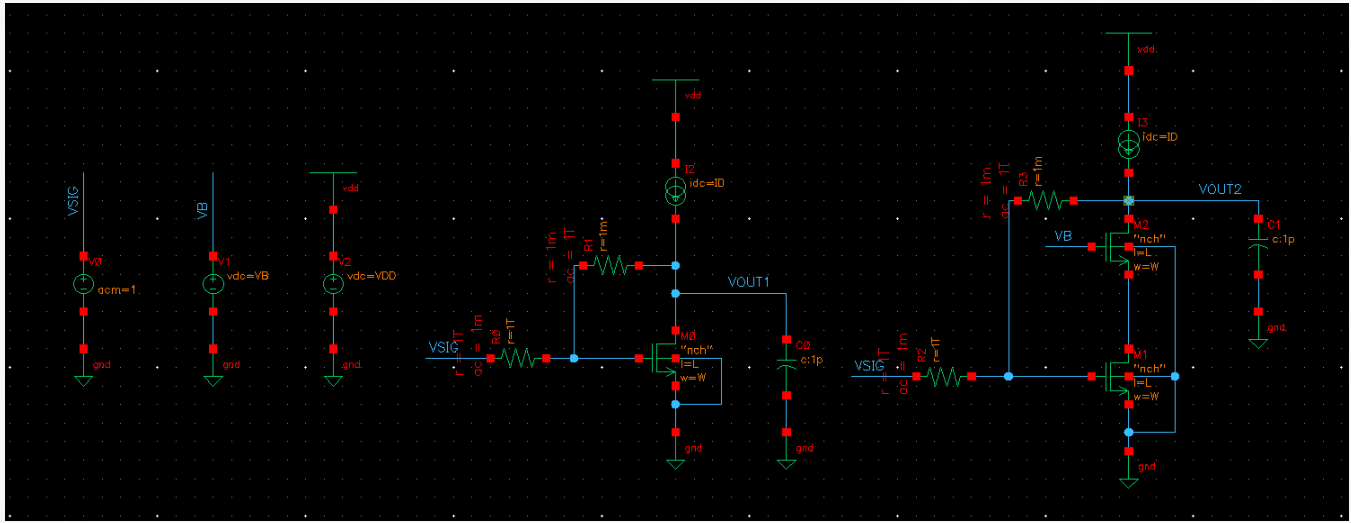
$\rightarrow L = 0.35\mu\text{m}$

$\rightarrow V_{DD} = 1.8\text{V}$

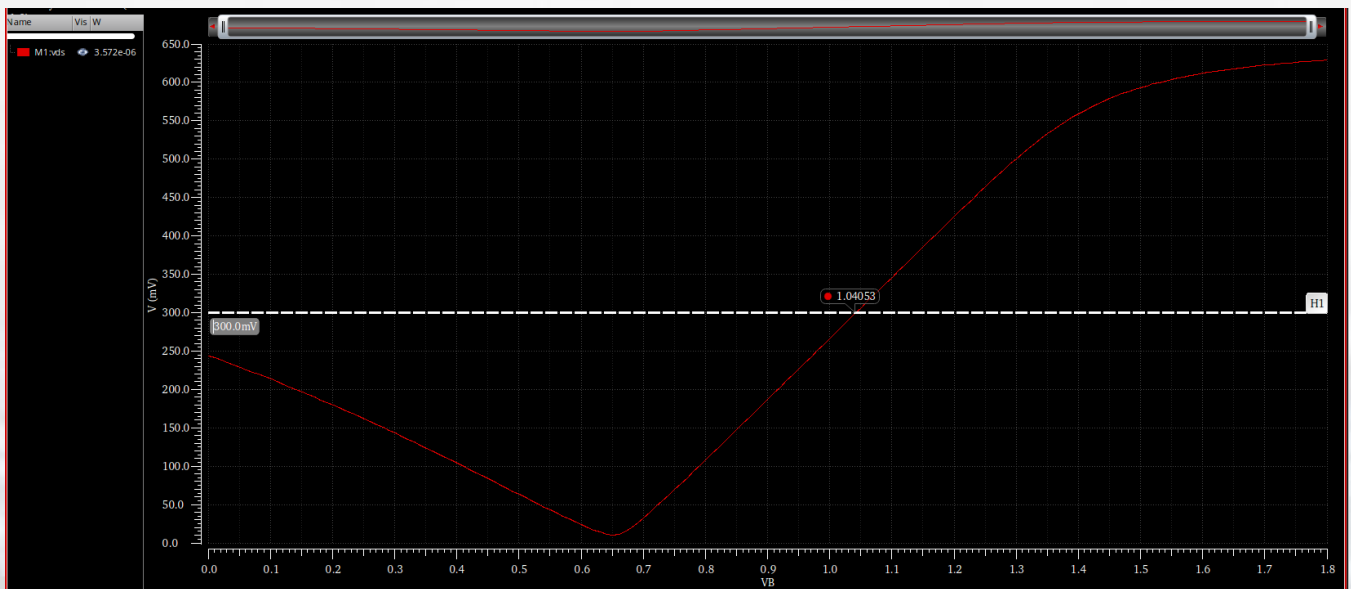
$\rightarrow V_{GS} = 659.041\text{mV}$

PART 2: Cascode for Gain

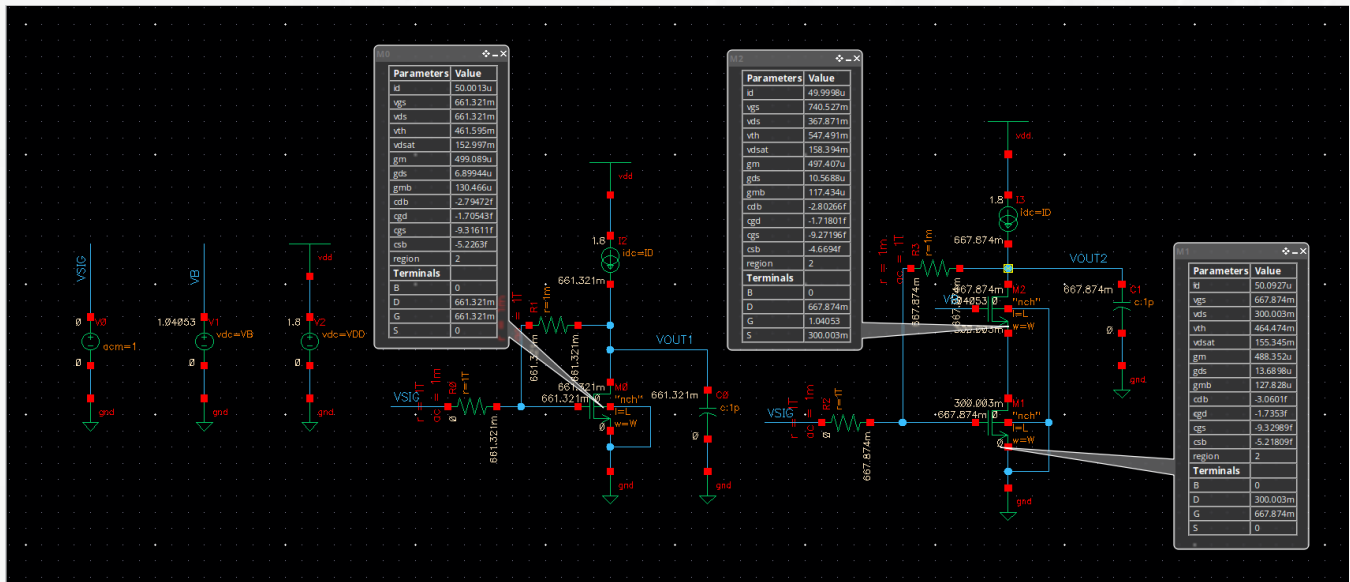
1. OP Analysis



Sweep V_B from 0 to 1.8v



The value of V_B at ($V_{DS} = V^* + 100\text{mV} = 300\text{mV}$) is 1.04v



- All transistors are in saturation region
- The two transistors which have its source's terminal connected to the ground have nearly similar V_{TH} (461.5mv and 464mv), but the other one has very difference V_{TH} value due to body effect.
- For all transistors in the circuit:

$$g_m \gg g_{ds}$$

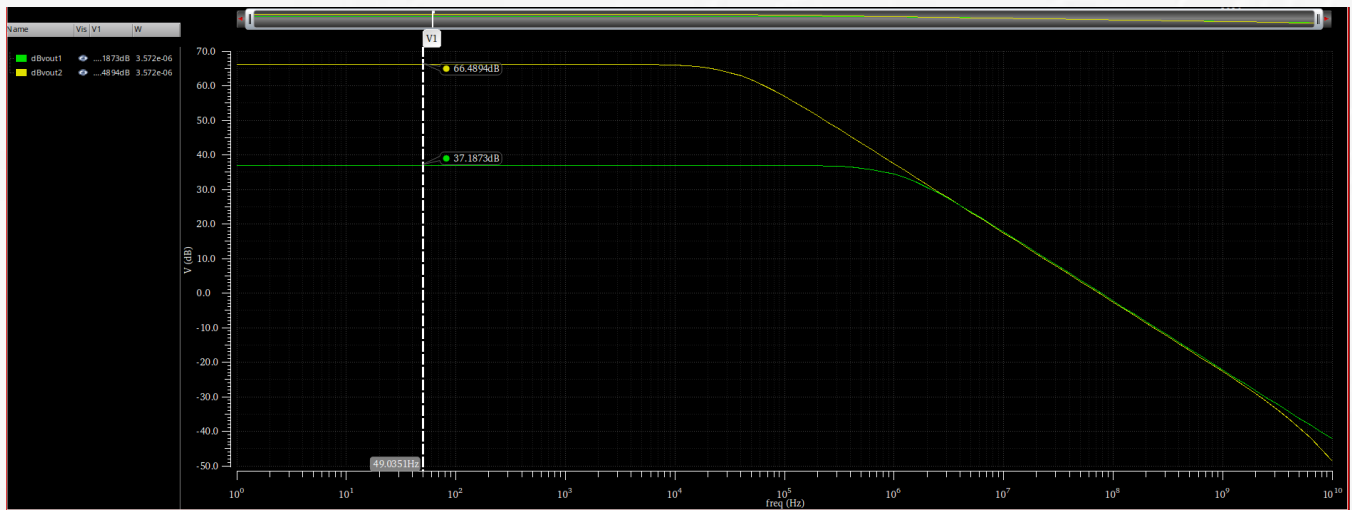
$$g_m > g_{mb}$$

$$|c_{gs}| > |c_{gd}|$$

$$|c_{sb}| > |c_{db}|$$

2. AC Analysis

output	Common source circuit	Cascode circuit
dB 20gain	37.19dB	66.49dB
DC gain	72.34	2111
Bandwidth BW	1.094MHz	35.89KHz
GBW	79.34MHz	75.94MHz
UGF	79.14MHz	76.31MHz



Bode plot magnitude

By hand analysis for the common source circuit (V_{out1}):

$$DC \text{ gain} = g_m r_o = \frac{g_m}{g_{ds}} = \frac{500\mu\Omega^{-1}}{6.9\mu\Omega^{-1}} = 72.46$$

$$BW = \frac{1}{2\pi r_o C_L} = \frac{1}{2\pi \frac{1}{6.9\mu\Omega^{-1}} 1pF} = 1.098MHz$$

$$GBW = DC \text{ gain} \times BW = 72.46 \times 1.098MHz = 79.56MHz$$

$$\text{let } w = 2\pi \times UGF \text{ rad/s} \rightarrow 1 = \frac{DC \text{ gain}}{\sqrt{1+(w r_o C_L)^2}} = \frac{72.46}{\sqrt{1+(2\pi \times UGF \times \frac{1}{6.9\mu\Omega^{-1}} 1pF)^2}} \rightarrow UGF = 79.56Hz$$

By hand analysis for the cascode circuit (V_{out2}):

$$R_{eq} \approx r_{o2}(g_{m2} + g_{mb2})r_{o1} = \frac{1}{10.57\mu\Omega^{-1}} (497.5\mu\Omega^{-1} + 117.5\mu\Omega^{-1}) \frac{1}{13.69\mu\Omega^{-1}} \approx 4.25M\Omega$$

$$DC \text{ gain} = g_{m1} R_{eq} = 488.35\mu\Omega^{-1} \times 4.25M\Omega \approx 2075.5$$

$$BW = \frac{1}{2\pi R_{eq} C_L} = \frac{1}{2\pi 4.25M\Omega \times 1pF} = 37.44KHz$$

$$GBW = DC \text{ gain} \times BW = 2075.5 \times 37.44KHz = 77.7MHz$$

$$\text{let } w = 2\pi \times UGF \text{ rad/s} \rightarrow 1 = \frac{DC \text{ gain}}{\sqrt{1+(w R_{eq} C_L)^2}} = \frac{2075.5}{\sqrt{1+(2\pi \times UGF \times 4.25M\Omega \times 1pF)^2}} \rightarrow UGF = 77.7Hz$$

parameter	Common source circuit			Cascode circuit		
	Simulation	By hand	Error	Simulation	By hand	Error
DC gain	72.34	72.46	0.12	2111	2075.5	35.5
BW	1.094MHz	1.098MHz	0.004MHz	35.89KHz	37.44KHz	1.55KHz
GBW	79.34MHz	79.56MHz	0.22MHz	75.94MHz	77.7MHz	1.76MHz
UGF	79.14MHz	79.56Hz	0.42MHz	76.31MHz	77.7MHz	1.39MHz

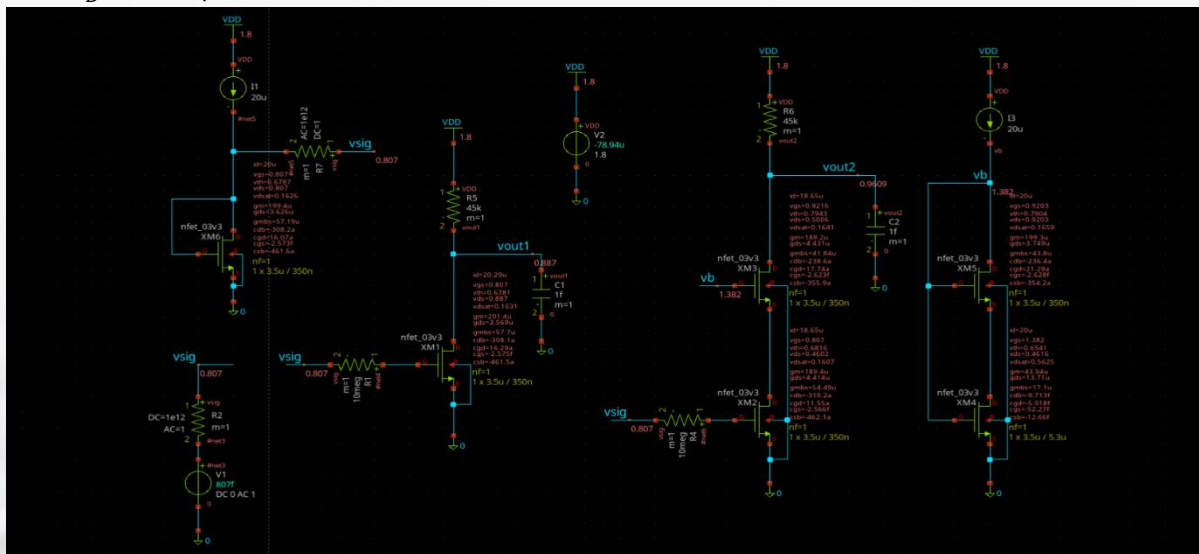
Comments on the results:

- The **gain** of the cascode amplifier is much larger than common source amplifier, but the common source amplifier has much larger **bandwidth**, and this is because the cascode amplifier has larger output resistance ($4.25M\Omega \gg 0.14M\Omega$). (note: $DC\ gain \propto R_{out}, BW \propto \frac{1}{R_{out}}$)
- GBW and UGF are **nearly similar** ($GBW1 \approx GBW2, UGF1 \approx UGF2$) for both circuits since $GBW = DC\ gain \times BW \propto \frac{R_{out}}{R_{out}} = 1$ and $UGF \approx \frac{DC\ gain}{2\pi R_{out}C_L} \propto \frac{R_{out}}{R_{out}} = 1$.
- Note that:** the values of the BW, GBW and UGF that getting from **hand analysis** are always larger than that getting from the **simulation** since in hand analysis we ignore the values of the parallel capacitance of the transistor.

PART 2: Cascode for BW

OP Analysis

- Calculate R_D analytically such that the voltage drop on it is $\approx \frac{V_{DD}}{2}$
- $R_D = \frac{V_{DD}}{2I_D} = \frac{1.8V}{2 \times 20\mu A} = 45K\Omega$



Operating point analysis results

- Check that all transistors operate in saturation. Does any transistor operate in triode? Why?
- All transistors are in saturation region except **M4** in **triode** region, **because** its $V_{GD} = V_{GS} - V_{DS} = 0.9204 > V_{TH}$ and $V_{DS} > V_{Dsat}$.

AC Analysis

- Use measure expressions to calculate parameters (DC gain, BW, GBW, and UGF) and export them to a text file.
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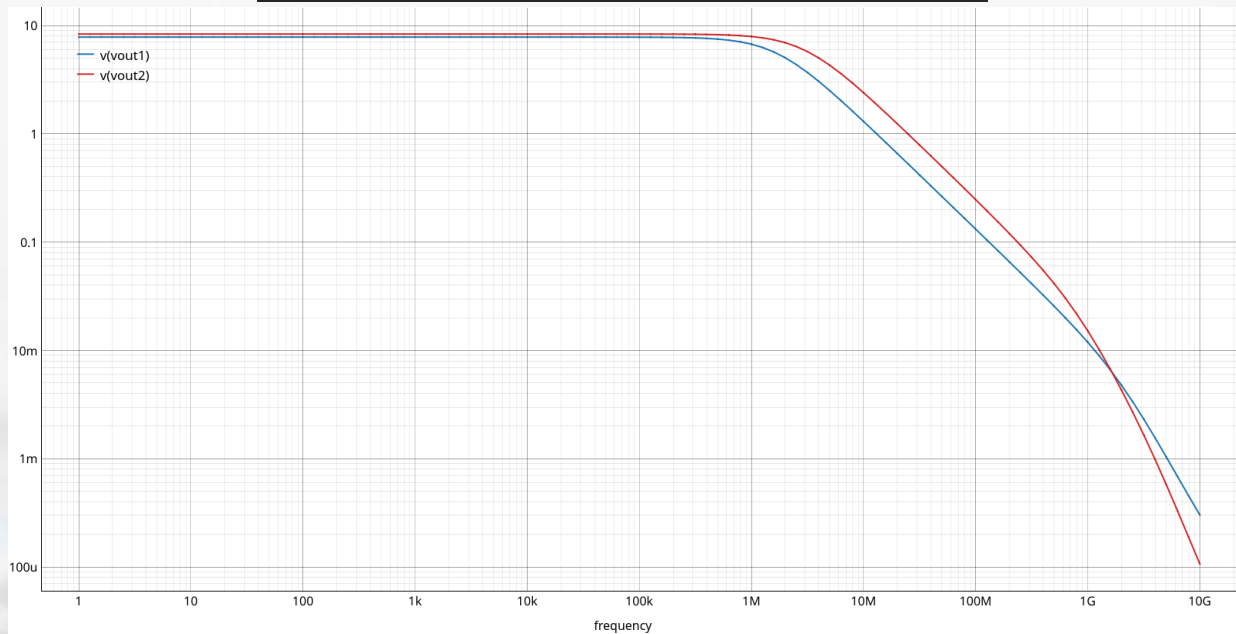
dc_gain_1 = 7.806966e+00
bw_1 = 1.694538e+06
gbw_1 = 1.322920e+07
ugf_1 = 1.318210e+07

```

```

dc_gain_2 = 8.336792e+00
bw_2 = 3.007130e+06
gbw_2 = 2.506982e+07
ugf_2 = 2.487425e+07

```



• Body plot: ■ cascode, ■ CS

- Using small signal parameters from OP simulation, perform hand analysis to calculate DC gain, BW, and GBW of both circuits.

- For CS: i/p pole dominant

$$A_v = g_m \left(\frac{1}{g_{ds}} // R_D \right) = 201.4 \mu S \left(\frac{1}{3.569 \mu S} // 45 K\Omega \right) \approx 7.8$$

$$BW = \frac{1}{2\pi R C_{in}} = \frac{1}{2\pi 10 M\Omega (1.7 fF (1+7.8) + 2.3 fF)} = 0.922 MHz$$

$$UGF = GBW = 7.8 \times 0.922 M = 7.74 \times 10^6$$

- For Cascode for BW: i/p pole dominant

$$A_v \approx g_m R_D = 189.4 \mu S 45 K\Omega \approx 8.523$$

$$BW = \frac{1}{2\pi R C_{in}} = \frac{1}{2\pi 10 M\Omega (2.335 fF + 2 \times 1.8 fF)} = 2.68 MHz$$

$$UGF = GBW = 8.523 \times 2.68 M = 22.84 \times 10^6$$

ID	20u
VGS	804.1m
L	350n
VDS	0.887
VSB	0
Stack	1
Results:	
Name	TT-27.0
29 csg	2.29f
30 cdg	1.695f

ID	18.65u
VGS	804.1m
L	350n
VDS	0.4602
VSB	0
Stack	1
Results:	
Name	TT-27.0
29 csg	2.335f
30 cdg	1.8f

parameter	Common source circuit			Cascode for BW circuit		
	Simulation	By hand	Error	Simulation	By hand	Error
DC gain	7.807	7.8	0.007	8.33	8.523	0.193
BW	1.69MHz	0.922MHz	0.768MHz	3MHz	2.68MHz	0.32MHz
GBW	13.2MHz	7.74MHz	5.46MHz	25.1MHz	22.84MHz	2.26MHz
UGF	13.18MHz	7.74Hz	5.44MHz	24.9MHz	22.84MHz	2.06MHz

- **Comment:**
- There is **large** error in CS circuit between simulation and hand analysis due to **miller approximation**, and **smaller** error in cascode circuit due to miller effect **significantly reduction**.
- The cascode has **higher BW** than the CS due to **miller effect reduction**.
- The cascode has **relatively higher DC gain** than the CS since $A_{v_{CS}} = g_m(R_D // r_o)$ and $A_{v_{cascd}} = g_m(R_D // R_{LFD})$ $R_{LFD} \gg r_o \rightarrow A_{v_{cascd}} > A_{v_{CS}}$.

